PYTHON ACTIVITY-BASED LEARNING

Case Study 08: Student Academic Risk Classification

Dr. RVR NRI Institute of Technology | Python Programming

Student Name: _____ **Roll No:** _____

Branch / Section: _____

Problem Statement

Classify students into academic risk categories using marks, attendance and laboratory scores.

Objectives

• Combine student marks, attendance and lab scores.

• Identify multiple risk factors.

• Classify students as Low, Medium or High risk.

Assumptions

• Marks and lab scores are out of 100; attendance is a percentage.

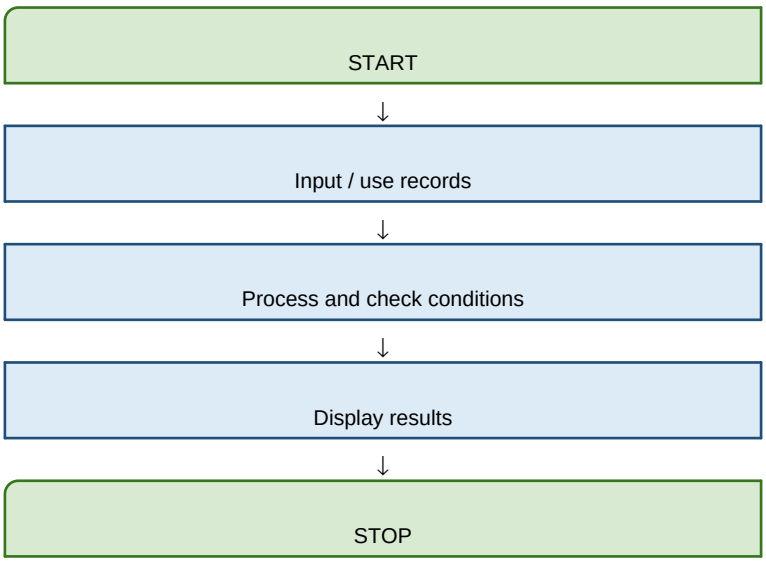
• High risk: marks below 40 or attendance below 60 or lab below 40.

• Medium risk: marks below 50 or attendance below 75 or lab below 50; others are low risk.

Algorithm

1. Store each student's name, marks, attendance and lab score in a tuple.
2. Unpack the tuple in a loop.
3. Apply high-risk conditions first.
4. Otherwise apply medium-risk conditions; all remaining students are low risk.
5. Display each student's category.

Flowchart



Pseudocode

```
START
STORE student records
FOR each record
    UNPACK name, marks, attendance, lab
    IF marks < 40 OR attendance < 60 OR lab < 40
        risk = HIGH
    ELSE IF marks < 50 OR attendance < 75 OR lab < 50
        risk = MEDIUM
    ELSE
        risk = LOW
    DISPLAY name and risk
STOP
```

Python Source Code

```
students = [
    ("Asha", 75, 90, 80),
    ("Bharat", 45, 70, 60),
    ("Charan", 35, 85, 70),
    ("Divya", 65, 55, 80)
]

for name, marks, attendance, lab in students:
    if marks < 40 or attendance < 60 or lab < 40:
        risk = "High"
    elif marks < 50 or attendance < 75 or lab < 50:
        risk = "Medium"
    else:
        risk = "Low"

    print(name, ":", risk, "Risk")
```

Test Cases

Test input / condition	Expected result
Asha: 75,90,80	Low Risk
Bharat: 45,70,60	Medium Risk
Charan: 35,85,70	High Risk
Divya: 65,55,80	High Risk

Output Screenshot

Sample console output (formatted to show the expected execution result):

```
Asha : Low Risk
Bharat : Medium Risk
Charan : High Risk
Divya : High Risk
```

Conclusion

The program combines academic indicators and uses conditional statements to classify student risk.